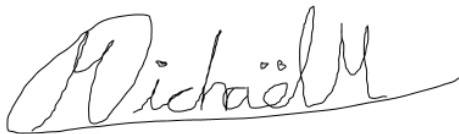


LAB 3: Dynamic Systems Modeling in Simscape

Course Number	MENG 3020
Course Title	System Modeling & Simulation
Semester/Year	Fall 2025

Lab/Tutorial Report No.	Lab 3
Report Title	Dynamic Systems Modeling in Simscape
Section No.	ONA
Group No.	N/A
Submission Date	October 19 th , 2025
Due Date	October 19th, 2025

Student Name	Signature*	Total Mark
Michael McCorkell		/ 50

* By signing above, you attest that you have contributed to this submission and confirm that all work you have contributed to this submission is your work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a ZERO on the work or possibly more severe penalties.

<https://academic-regulations.humber.ca/2021-2022/17.0-ACADEMIC-MISCONDUCT>

LAB 3 Grading Sheet

Student First Name: Michael	Student Last Name: McCorkell
Part A.1: Build the Model, Set the Component Parameters & Run the Simulation	/5
Part A.2: Current Circulation in the Active Phase & Freewheel Phase	/10
Part A.3: Effect of the Duty-Cycle on the Motor Voltage & Motor Speed	/10
Part A.4: Effect of the Rotor Inertia and Armature Inductance on the Motor Speed	/10
Post Lab Assignment	/10
General Formatting: Clarity, Writing style, Grammar, Spelling, Layout of the report	/5
Total Mark	/50

LAB 3: Dynamic Systems Modeling in Simscape

OBJECTIVES

- To model and simulate a buck converter system in Simscape to speed control of a DC motor.
- To understand the circulation of the current in the circuit in active phase and in freewheel phase.
- To visualize and evaluate the effect of the duty cycle, switching frequency and the inductance value on the current, voltage and speed of DC motor.

Simscape Model of DC Motor Control by Buck Converter

Introduction

A **Buck Converter** is a DC-to-DC power converter, which steps down voltage from its input (source) to output (load). It typically contains two semiconductor switches, a **diode**, and a **transistor**, to control the direction of the current in the circuit. Buck converters enable us to obtain an adjustable DC voltage from a fixed DC voltage, which is particularly used to **speed control** of **DC motors**. Figure 1 shows a typical configuration of a simple buck converter.

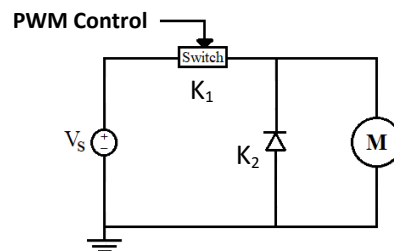


Figure 1: Typical Buck Converter

V_s is a fixed **DC voltage source**. **M** is a **brushed DC motor**. Switch K_1 is a **power transistor** which can be turned ON and OFF by a PWM control signal. Switch K_2 is a **diode**, which is called flyback or freewheeling diode.

A PWM controlled buck converter has two phases of operation: **Active phase** and **Freewheel phase**.

In the **active phase**, switch K_1 is **ON**, and K_2 is **OFF**.

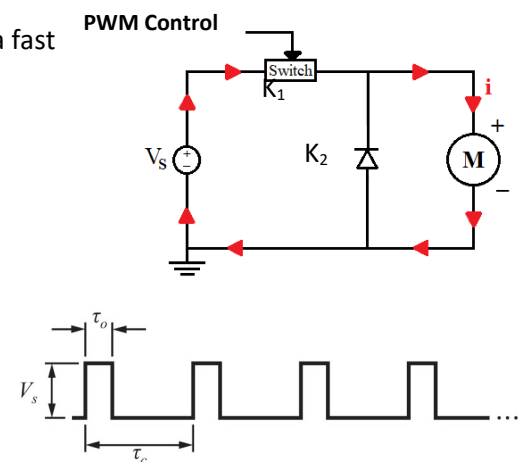
- The current that passes through the motor is identical to that passes through the supply.
- The current in the diode branch is zero.
- The high frequency PWM signal turns ON and OFF the K_1 switch at a fast rate, which allows us to adjust the average or effective value of the applied voltage to the DC motor.
- The average voltage depends on the duty cycle of the PWM signal.
- Duty cycle is the amount of time the PWM signal is ON versus the amount of time the signal is OFF in a single period.

$$\text{Duty - cycle} = \frac{\tau_o}{\tau_c}$$

- The effective applied voltage V_e is determined as:

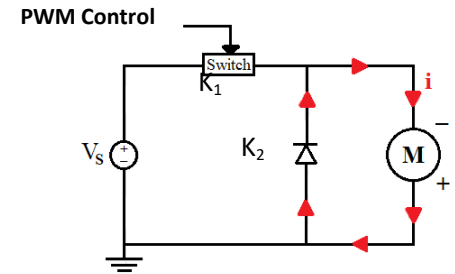
$$V_e = V_s \frac{\tau_o}{\tau_c}$$

which increases by increasing the duty-cycle of the PWM signal.



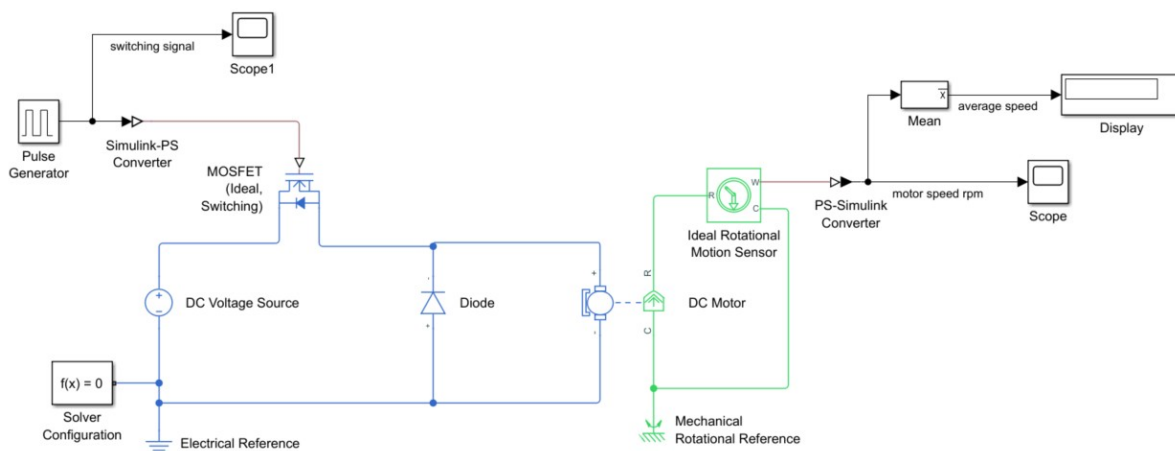
In the **freewheel phase**, switch K_1 is **OFF**, and K_2 is **ON**.

- The current that passes through the motor is identical to that passes through the diode branch.
- The current in the supply is zero.
- Turning OFF the switch K_1 does not immediately stop current in the motor windings.
- The inductive behavior of the DC motor causes current to continue to flow when the switch K_1 is opened suddenly.
- Adding a flyback diode in parallel with the motor provides a path for dissipation of stored energy when the switch K_1 is OFF.

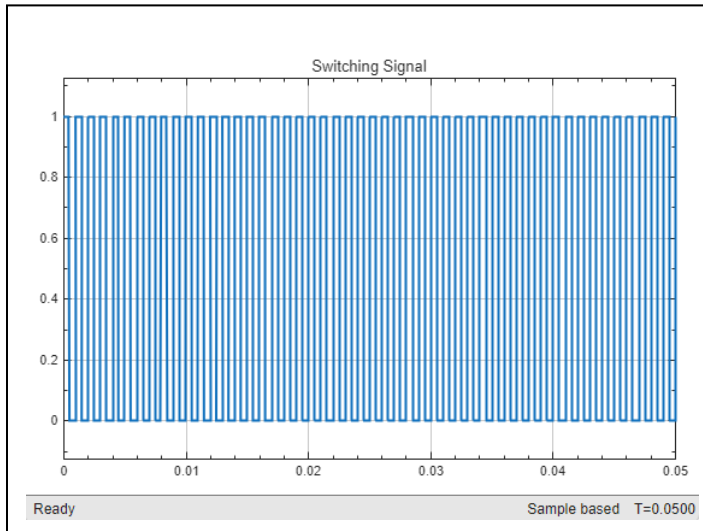


A.1. Build the Model, Set the Component Parameters & Run the Simulation

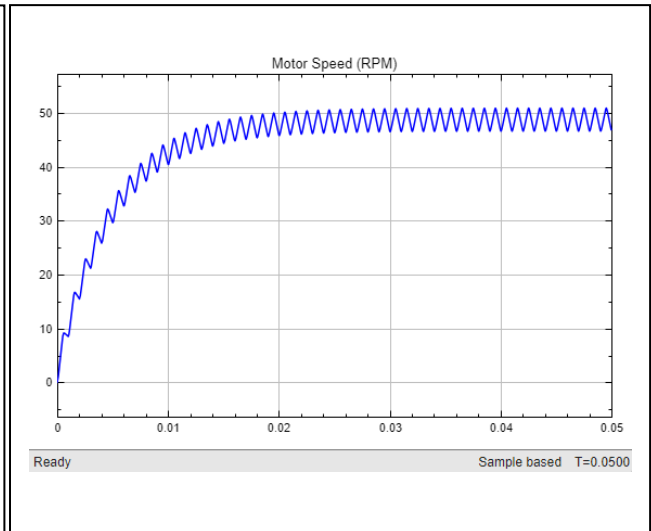
1. Open a new **Simulink** model page. Find and place the following elements and components from the **Simulink Library Browser**.
 - Simscape > Foundation Library > Electrical > Electrical Elements > Diode
 - Simscape > Foundation Library > Electrical > Electrical Elements > Electrical Reference
 - Simscape > Foundation Library > Electrical > Electrical Sources > DC Voltage Source
 - Simscape > Foundation Library > Mechanical > Rotational Elements > Mechanical Rotational Reference
 - Simscape > Foundation Library > Mechanical > Mechanical Sensors > Ideal Rotational Motion Sensor
 - Simscape > Electrical > Semiconductors & Converters > MOSFET (Ideal Switching)
 - Simscape > Electrical > Electromechanical > Brushed Motors > DC Motor
 - Simscape > Electrical > Specialized Power Systems > Sensors and Measurements > Mean
 - Simscape > Utilities > PS-Simulink Converter
 - Simscape > Utilities > Simulink-PS Converter
 - Simscape > Utilities > Solver Configuration
 - Simulink > Sinks > Scope
 - Simulink > Sinks > Display
 - Simulink > Sources > Pulse Generator
2. Arrange and connect the blocks as shown below. The model involves the **electrical field** and the **mechanical rotational field**:



Switching PWM Signal



DC Motor Speed (rpm)



7. Since the PWM switching frequency is very fast and the load inertia is very low, the commutation frequency causes ripple in the motor speed. What is the **average final speed** of the motor?

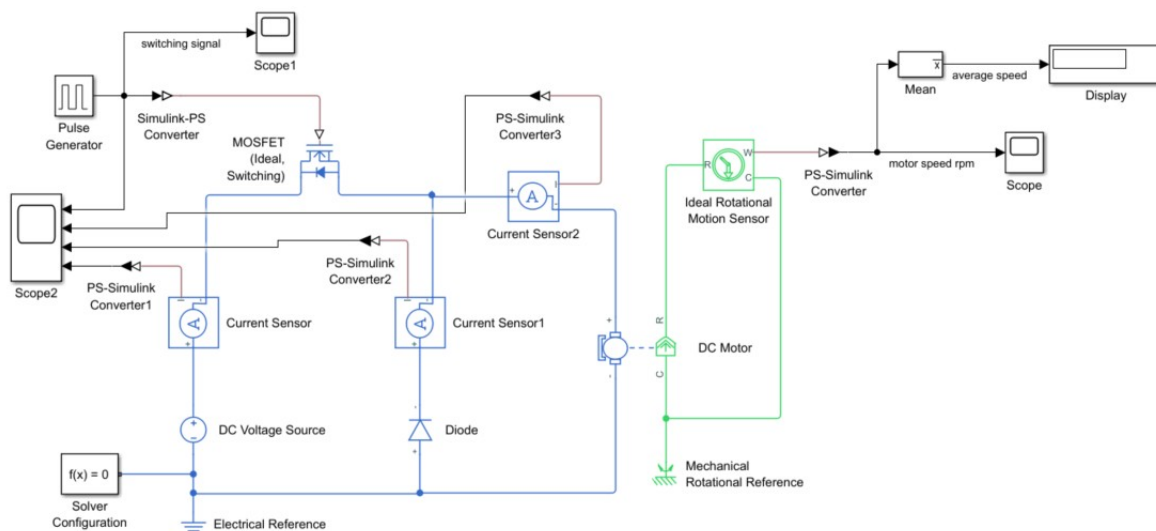
48.72 RPM

A.2. Current Circulation in the Active Phase & Freewheel Phase

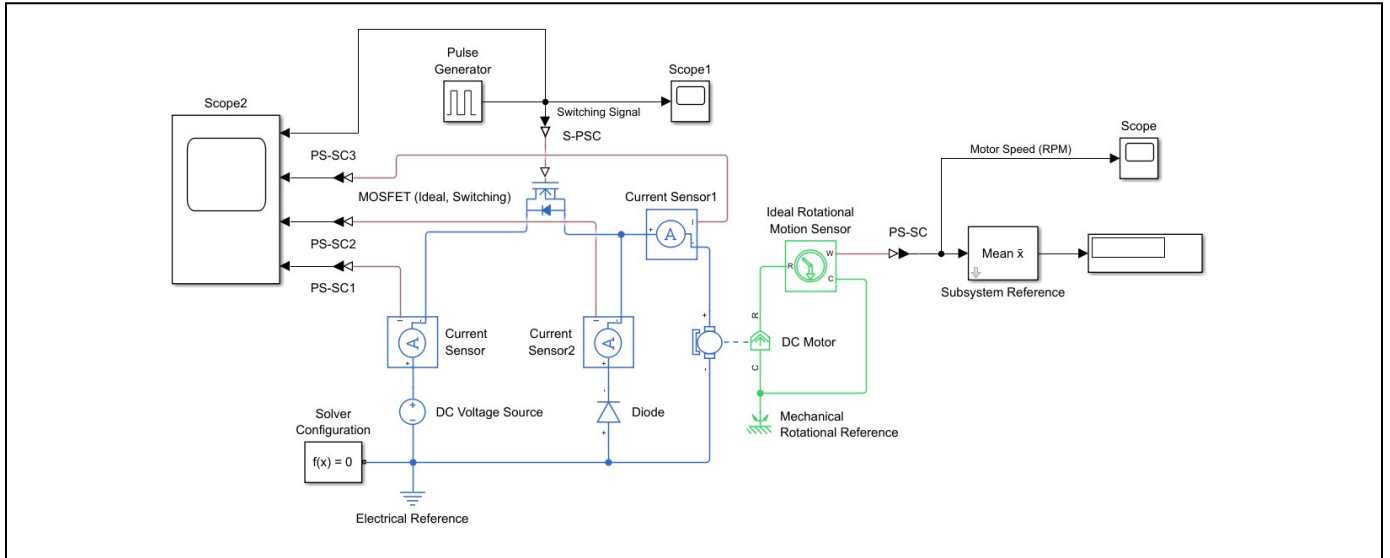
To view the current circulation in the buck converter, we will modify the **Simscape** model by adding the current sensors in each of the circuit branches.

8. Select and place the following blocks from the library and modify the circuit diagram as shown below:

- **Simscape > Foundation Library > Electrical > Electrical Sensors > Current Sensor**
- **Simscape > Utilities > PS-Simulink Converter**
- **Simulink > Sinks > Scope**

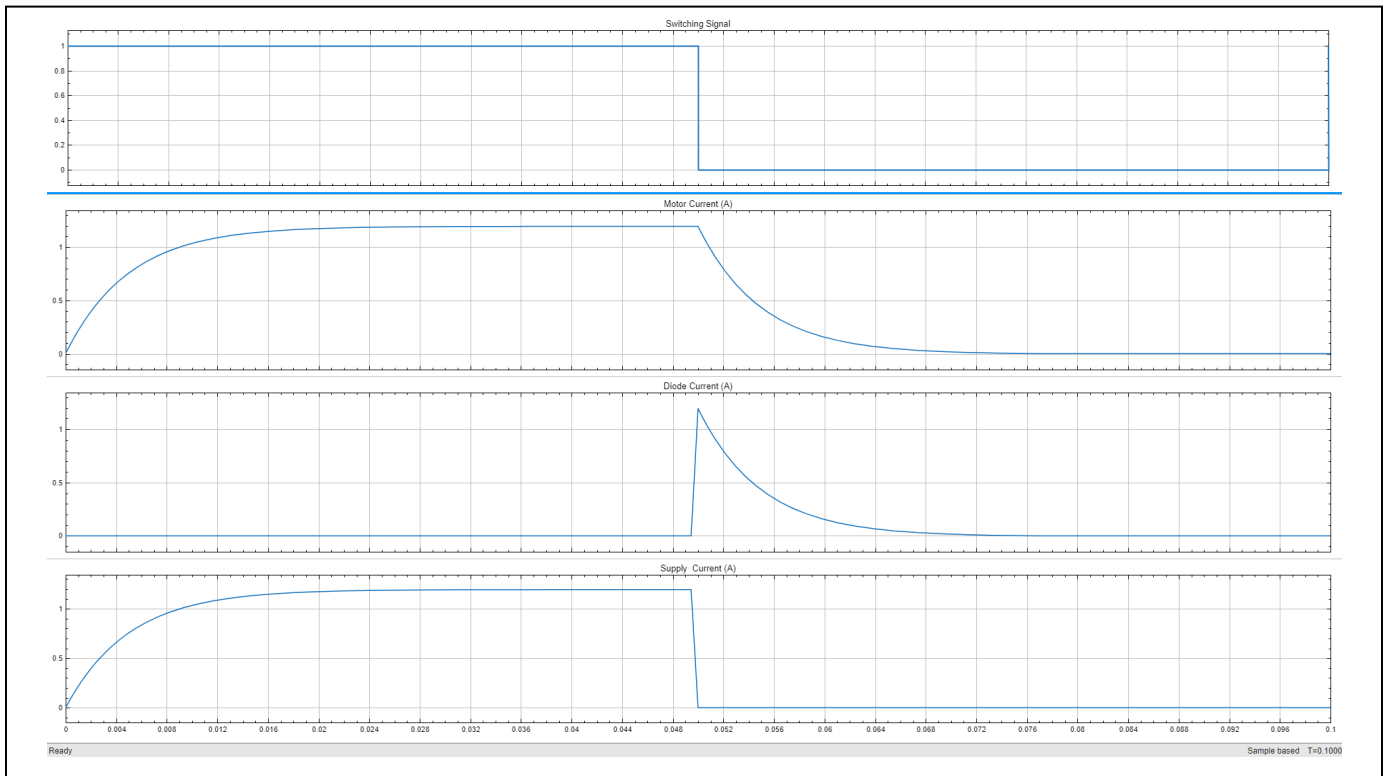


9. Provide your **Simscape** model diagram below:



10. To better understanding the current flow, we set the frequency of the switching signal in **Pulse Generator** to **10Hz**.
11. Adjust the **Fundamental frequency** in **Mean** block to **10Hz** to match with the switching frequency.
12. Set the simulation **Stop Time** to **0.1 sec** and **Run** the simulation.
13. Open the **Scope** block to view the currents waveform of each branch in the circuit.
14. Provide the **currents waveform** graph below. Identify the **active phase** and **freewheel phase** on the graph.

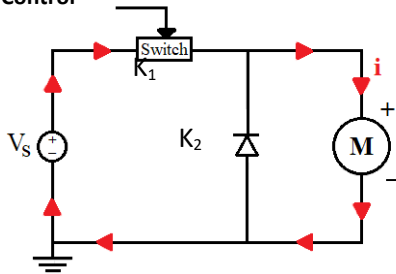
Currents Waveform



15. Briefly describe the **current waveforms** and the **motor speed** in the active phase and the freewheel phase.

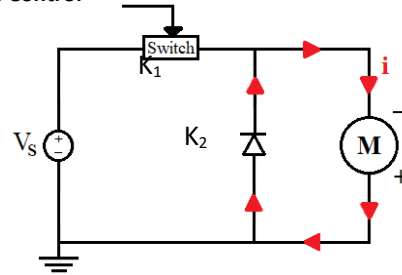
In the active phase, both motor speed and current increase until they reach their maximum. During the freewheeling phase, the diode provides a path for the motor current, allowing it to gradually decay rather than stopping abruptly, so the motor continues to coast while the current waveform slowly decreases.

PWM Control



Active Phase

PWM Control



Freewheel Phase

16. To check the effect of the switching frequency on the current waveforms, set the switching frequency to the given values in **Table 1**. Observe the **current waveforms** and collect the **average motor speed** data. Make sure to change the **Fundamental frequency** of the **Mean** block based on the assigned switching frequency.

Table 1

Switching Frequency	10 Hz	20 Hz	40 Hz	80 Hz	200 Hz	500 Hz
Average Speed (rpm)	48.91	48.79	48.76	48.74	48.73	48.72

17. Briefly describe the effect of increasing the switching frequency on the **current ripples** and the **motor speed**.

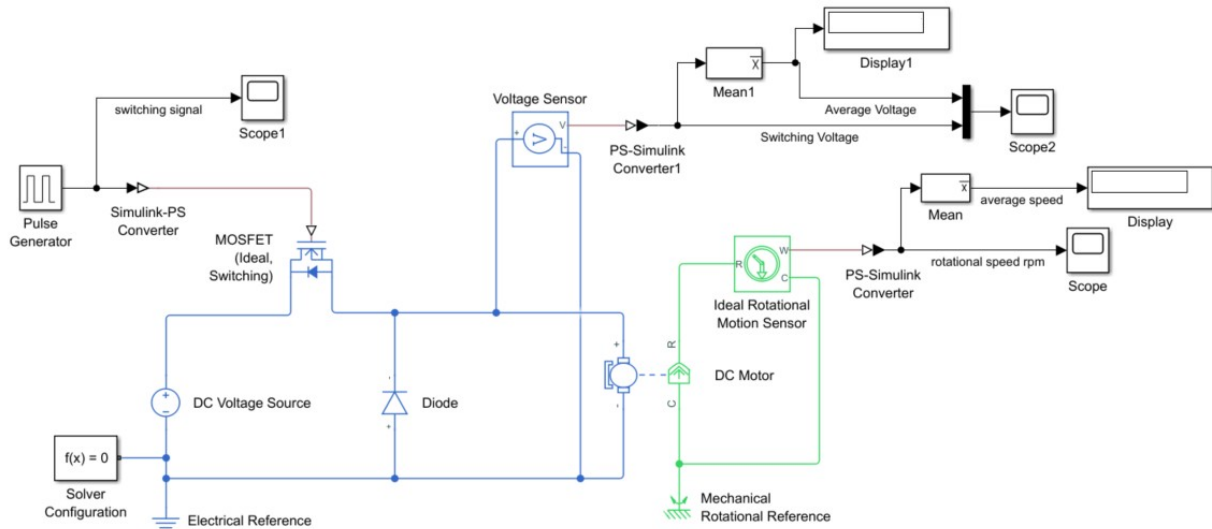
Increasing the switching frequency slightly reduces the current ripples in the motor, as the voltage is applied in shorter, more frequent pulses, allowing the motor's inductance to smooth the current more effectively. However, the average motor speed shows only a minimal decrease with higher switching frequencies, as seen in the table, indicating that the primary effect of increasing frequency is improved current stability rather than significant changes in speed.

A.3. Effect of the Duty-Cycle on the Motor Voltage & Motor Speed

In this step the goal is to observe the variation of the average applied voltage and speed of the DC motor by changing the duty-cycle of the PWM signal.

18. Modify the **Simscape** model from previous part by adding the following components as shown below:

- **Simscape > Foundation Library > Electrical > Electrical Sensors > Voltage Sensor**
- **Simscape > Electrical > Specialized Power Systems > Sensors and Measurements > Mean**
- **Simscape > Utilities > PS-Simulink Converter**
- **Simulink > Sinks > Scope**

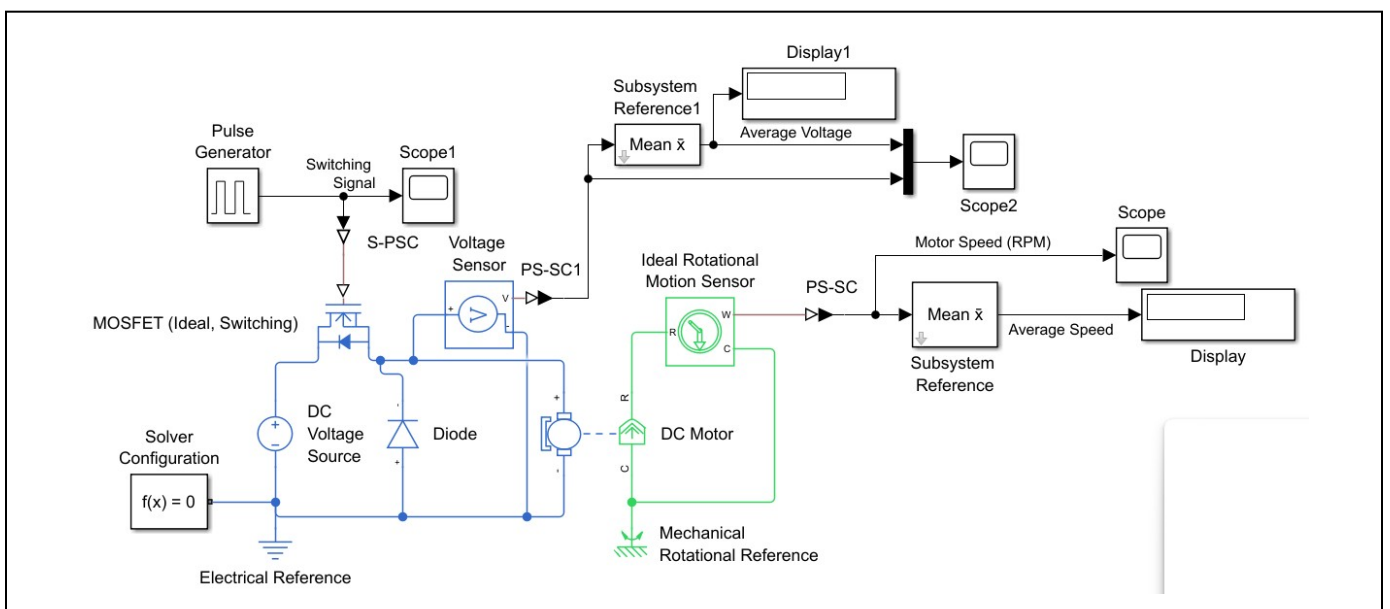


19. Set back the **Pulse Generator** parameters to the give original values.

Amplitude = 1, Period = 0.001 sec, Pulse Width = 50 percent, Phase delay = 0 second

20. Set the **Fundamental frequency** of the **Mean** blocks to **1000 Hz**.

21. Provide your **Simscape** model diagram below:



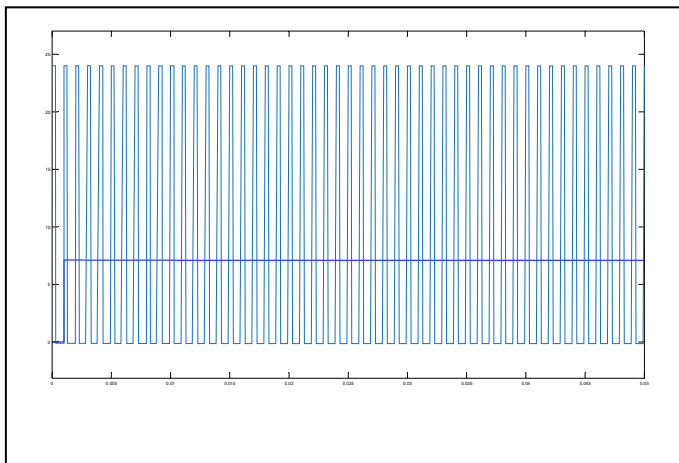
22. Change the **duty-cycle** of the PWM switching signal in the **Pulse Generator** block for each of the given values in **Table 2**. Run the simulation for **0.05sec** in each case. Provide the required graphs, calculations, and data from the simulation and complete **Table 2**.

Table 2

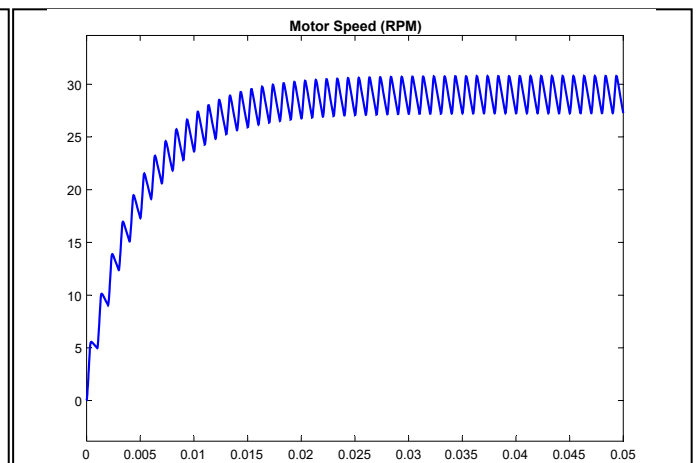
Duty-Cycle	10%	30%	50%	70%	90%	99%
Average Speed (rpm)	9.401 RPM	29.04 RPM	48.72 RPM	68.43 RPM	88.18 RPM	97.08 RPM
Average Voltage (V)	2.3 V	7.104 V	11.92 V	16.74 V	21.57 V	23.75 V

Provide required samples of simulation graphs below:

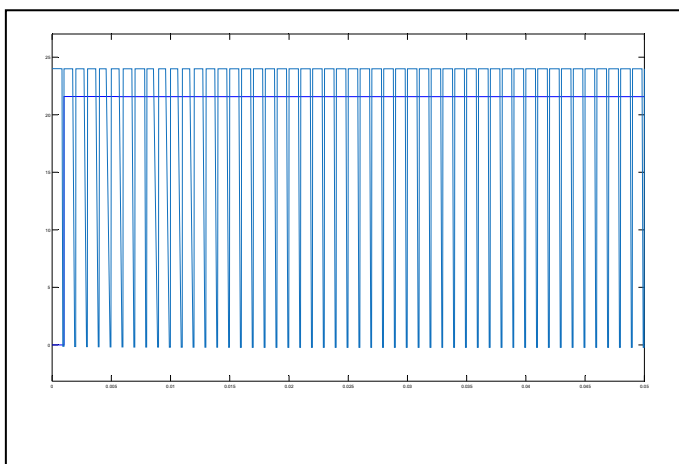
Applied Voltage (V) (Duty-cycle = 30%)



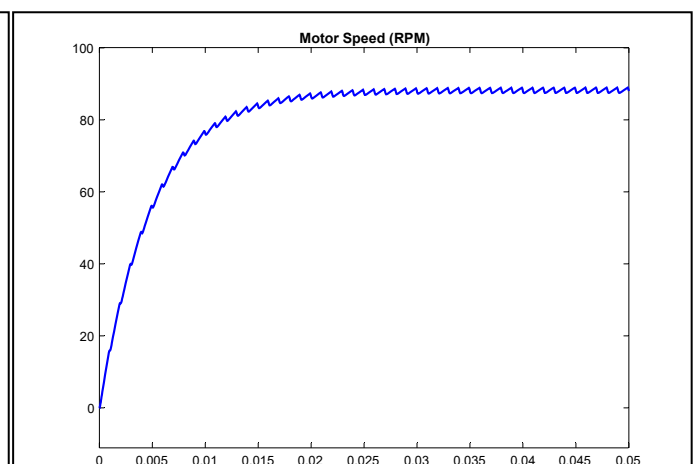
DC Motor Speed (rpm) (Duty-cycle = 30%)



Applied Voltage (V) (Duty-cycle = 90%)



DC Motor Speed (rpm) (Duty-cycle = 90%)



23. Briefly describe the effect of increasing the duty-cycle on the ripples, the average motor speed, and the average voltage applied to the motor.

As the duty cycle of a PWM signal increases, the average voltage applied to the motor rises, causing the motor to spin faster and increasing its average speed. At the same time, the ripples in voltage and speed are reduced, resulting in smoother and more stable operation. In this way, a higher duty cycle not only boosts the motor's performance but also improves its consistency.

A.4. Effect of the Rotor Inertia and Armature Inductance on the Motor Speed

In this part, the goal is to observe the effect of the **rotor inertia** and the **armature inductance** on the **average motor speed** and the **ripple**.

24. Set back the **Pulse Generator** parameters to the original values.

Amplitude = 1, Period = 0.001 sec, Pulse Width = 50 percent, Phase delay = 0 second

25. *Double-click* on the **DC Motor** block and change the **Rotor inertia** first to **10 g.cm²** and then to **0.001 g.cm²**

26. **Run** the simulation for **0.05sec** and open the **Scope** block and check the motor speed for each case. Provide the speed graphs and the average final speed below.

DC Motor Speed (Rotor Inertia = 10 g.cm²)

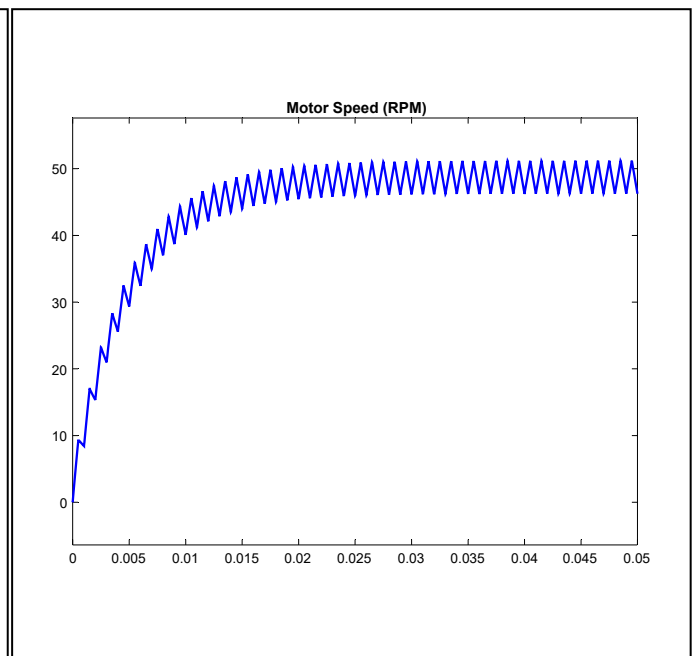
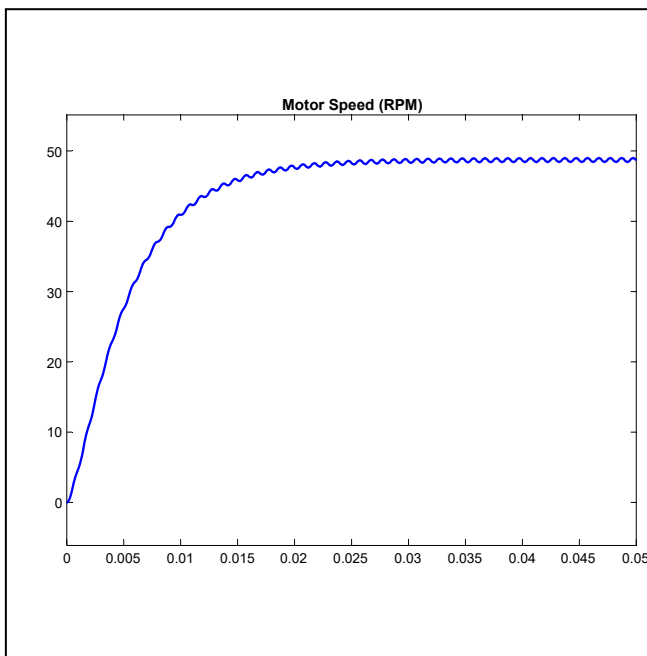
Average Final Speed (rpm):

48072

DC Motor Speed (Rotor Inertia = 0.001 g.cm²)

Average Final Speed (rpm):

48.72



27. Explain the effect of variation of rotor inertia on the ripple, time constant and the average speed of the motor.

When the rotor inertia was adjusted to 10, the ripples decreased, but the time constant and average speed stayed unchanged. Conversely, reducing the rotor inertia to 0.001 caused an increase in ripples, while the time constant and average speed remained unaffected.

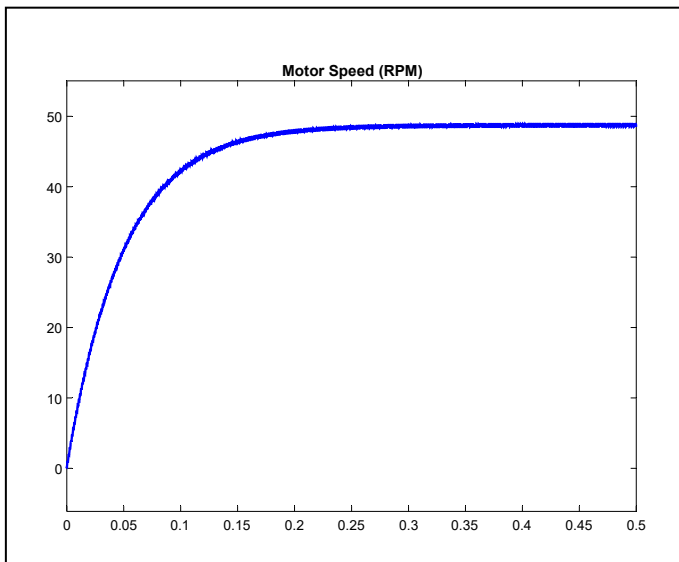
28. Set back the **Rotor inertia** to the original value of 0.5 g.cm^2 .
 29. *Double-click* on the **DC Motor** block and change the **Armature inductance** first to **1 H** and then to **10 mH**.
 30. **Run** the simulation and open the **Scope** block and check the motor speed for each case. Provide the speed graphs below.

NOTE: For $L_a = 1 \text{ H}$ set the simulation time to **0.5 sec** and for $L_a = 10 \text{ mH}$ set the simulation time to **0.02 sec** to capture the full speed response graph.

DC Motor Speed ($L_a = 1 \text{ H}$)

Average Final Speed (rpm):

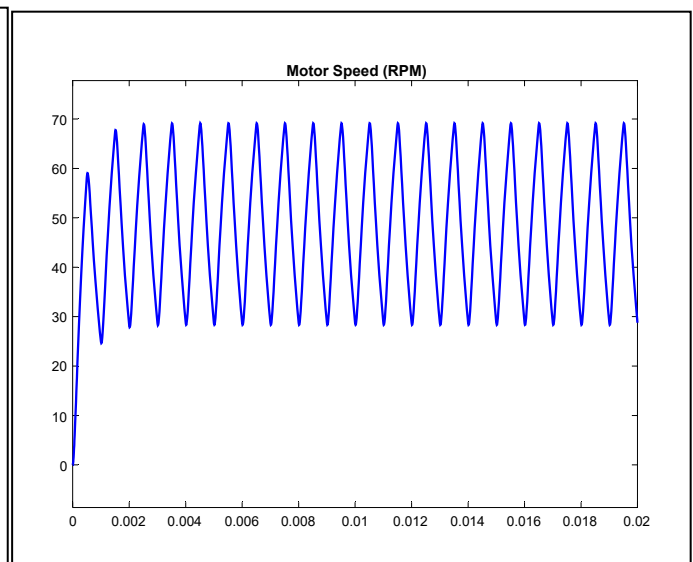
48.72



DC Motor Speed ($L_a = 10 \text{ mH}$)

Average Final Speed (rpm):

48.73



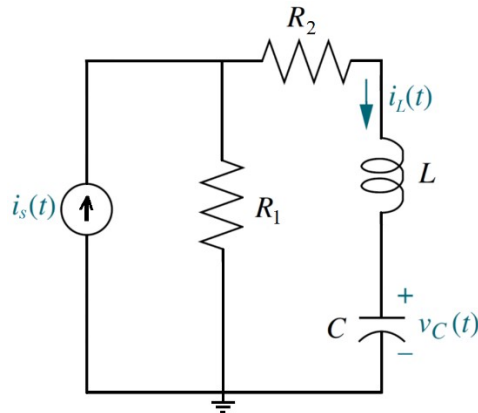
31. Explain the effect of changing the Armature inductance on the ripple, time constant and the average speed of the motor.

Setting the armature inductance to 1 H considerably eliminates ripples, decreases the time constant, and keeps the average speed virtually constant. When the inductance is increased to 10 mH, the ripples rise significantly, the time constant accelerates, but the average speed remains essentially constant.

Post Lab Assignment

1. Build the Simscape physical diagram for the following electric circuit.

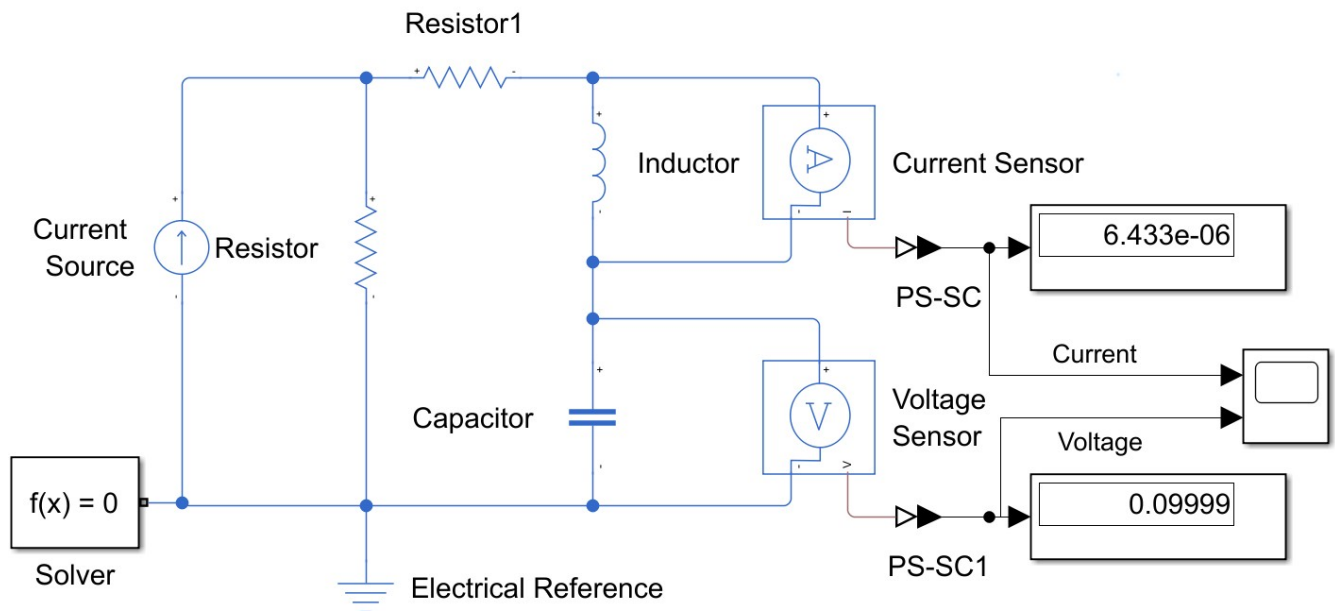
$$R_1 = 0.2 \, \Omega, \quad R_2 = 1 \, \Omega, \quad C = 0.01 \, F, \quad L = 0.1 \, H$$



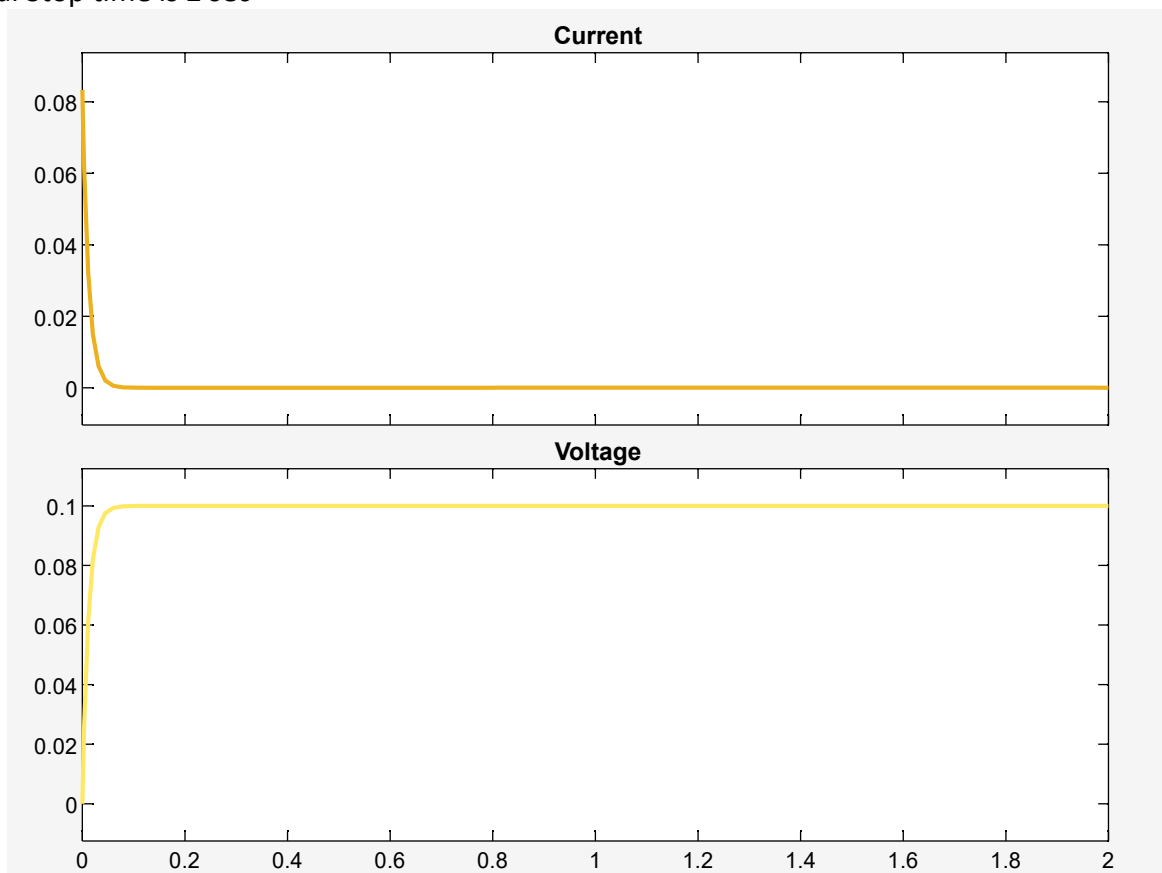
- Provide your Simscape model diagram.
- Select an appropriate simulation time and plot the voltage across the capacitor $v_c(t)$ and current through the inductor $i_L(t)$ if the input signal is $i_s(t) = 0.5 \, A$ and all initial conditions are $v_c(0) = 0$ and $i_L(0) = 0$. Provide the graphs.
- What is the final voltage of the capacitor? What is the final current of the inductor?
- Set the initial conditions to $v_c(0) = 2 \, V$ and $i_L(0) = 0.5 \, A$. Plot the voltage $v_c(t)$ and current $i_L(t)$ graphs. Provide the graphs?
- What is the final voltage of the capacitor? What is the final current of the inductor?

Post Lab)

a)



b) The final Stop time is 2 sec



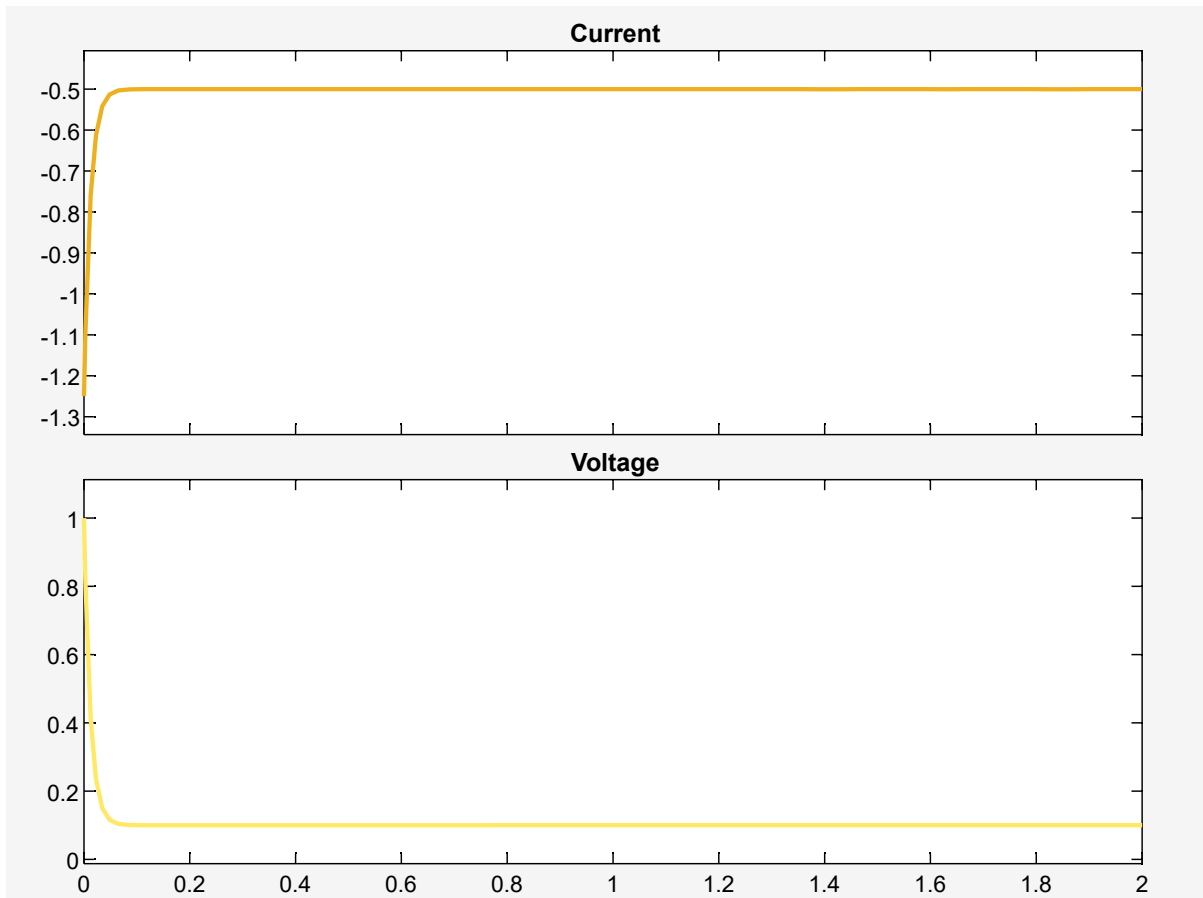
C)

Final Value

Induction (L): 0 A

Capacitor (C): 0.1 V

D)



E)

Final Value

Induction (A): -0.5001 A

Capacitor (V): 0.1001 V